Shot on motorola

2022-26

Full Marks : 70

Time : 3 hours

Answer from **both** the Groups as directed.

The figures in the right-hand margin indicate marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP—A

Answer any *four* questions. 10 × 4

1. Define data structure. Explain different types of data structure.

2. Define linked list. Write a C program for array implementation of a singly linked list.

3. Define Stack. Explain how to evaluate arithmetic expression using stack.

4. What is Queue ? Write a C function to perform insertion and deletion on Queue.

5. Define Binary Search Tree (BST). Write an algorithm to construct a BST.

6. What is a traversal of a Binary tree ? Construct the Binary tree with the following tree traversal.

   Inorder: *4, 8, 2, 5, 1, 6, 3, 7*

   Postorder: *8, 4, 5, 2, 6, 7, 3, 1*

7. Define AVL tree. Construct an AVL tree with the following data.

   21, 26, 30, 9, 4, 14, 28, 18

8. What is Merge sort ? Explain the procedure of merge sort with the following data

   6, 2, 11, 7, 4, 3, 19, 8

GROUP—B

Answer *all* the questions. 3 × 10

9. What are the limitations of arrays ?

10. Define polish notation.

11. What is Big-oh notation ?

12. Compare between linear search and binary search.

13. What is complete binary tree ?

14. Differentiate between Graph and Tree.

15. Write about the sparse array.

16. What is a spanning tree ?

17. What is Hashing ?

18. Define Circular Queue.

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2023(New)

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any **four** questions of the following : 10×4 = 40

1. Compare Linear search and Binary search. Write an algorithm for Binary search in an array.

2. Explain classification of data structure and operation on data structure.

3. What is linked list ? Explain different types of linked list. Write an algorithm to insert a node at the beginning of a singly linked list.

4. Write a C function to perform insertion and deletion operation on stack.

5. (a) Write an algorithm to delete an element from queue.

   (b) Write an algorithm to evaluate postfix expression.

6. Differentiate between Internal and External Sorting. Write an algorithm to implement Quick sort.

7. Write an algorithm for the following :

   (a) Inorder tree traversal

   (b) Preorder tree traversal

8. Construct an AVL tree with the following data :

   21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3, 7.

Group – B

Answer **all** questions : 3×10 = 30

9. Define data structure.

10. Mention the disadvantages of an array.

11. Define Sparse array.

12. Differentiate between Stacks and Queues.

13. What is Circular Queue ?

14. Differentiate between Binary Tree and Complete Binary Tree.

15. Define Big-oh notation.

16. What is a degree of a tree ?

17. What is the time complexity of a linear search ?

18. What is Priority Queue ?



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UG — BCA
(C – 3001)

2022

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any **four** questions of the following : 10×4 = 40

1. What is Data Structure ? Explain different data structure.

2. What is Sparse Array ? Explain 3-tuple representation of sparse matrix with the help of suitable example.

3. How does a linear linked list compare with linear array ? Write function to add a node at the beginning of linked list.

4. Translate this expression into postfix notation and then evaluate it :

$$12/(7-3)+2^{*}(3+8)-7$$

5. For a binary tree T, the in-order and post-order traversal sequences are the following :

In-order D C K E A H B Q J I

Post-order D K E C H Q J I B A

Draw the binary tree T.

6. Write a program to implement the concept of Stack.

7. Illustrate the working of the Bubble sort method of an array.

35 20 40 100 3 10 15

8. What is Binary Search Technique ? Write an algorithm / function of Binary Search Technique.

Group – B

Answer **all** question of the following : 3×10 = 30

9. What is enumerated data type ?

10. What is the criterion that every algorithm must satisfy ?

11. Define Polish notation.

12. What is Circular Linked List ?

13. Differentiate between STACK and QUEUE data structure.

14. Define height of a tree.

15. Briefly explain AVL-tree.

16. Write two application of Stack.

17. What is M-Way Merge Srot ?

18. Define DEQUEUE.

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2018

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer from both the Sections as directed.

Section – A

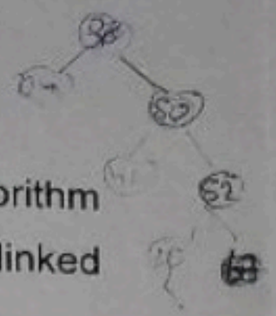
Answer any **four** questions : 10×4 = 40

1. What is a Sparse Matrix ? What are the advantages and disadvantages of sparse matrix representation ?

2. Write an algorithm to implement Heap Sort. Write the steps to sort the following elements by heap sort method :

   7, 28, 4, 72, 22, 39.

3. Make a binary search tree by considering the following numbers :



50, 24, 38, 24, 67, 40, 60, 52

4. What is a doubly linked list ? Write an algorithm to create and display node in a doubly linked list.

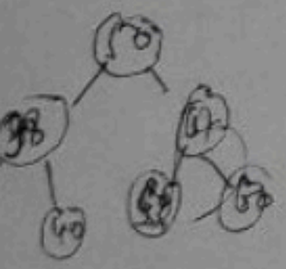
5. What is a Priority Queue ? Write an algorithm for insertion and deletion in a Priority Queue.

6. Show step-by-step construction of B-tree (t = 3) from the insertion of the following keys :

A, S, T, M, U, K, L, B, G, N

7. Compare and contrast linear search and binary search techniques.

8. Construct a binary tree using the following Pre-order and In-order traversals :



Pre-Order	In-Order
A	B
B	H
G	M

Pre-Order	In-Order
H	G
M	A
C	D
D	F
E	E
F	C

Section – B

Answer **all** questions : 3×10 = 30

9. Define Data Structures. Discuss the various types of data structures.

10. Explain row major representation of an array.

11. Differentiate between a stack and a queue.

12. Convert the following infix expression to postfix :

A * B – C + D / (E * F) + G

13. What is Hashing ?

14. Write the drawbacks of DQUEUES.

15. What do you mean by depth of tree ?

16. Discuss Merge Sort Algorithm.

17. What are the applications of stack ?

18. Define header node in linked list.

